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SLOTTED AIRFOIL :- FABRICATION & ANALYSIS

CourseName:

AerodynamicsII

CourseCode:

ASE 204

SubmittedTo:

Dr.BalajiRavi

SubmittedBy:

LAXMIPRIYA MURMU- 12208017

GOWTHAM KUMAR CHUKKA 12307541

GOWTHAM KUMAR TALLA- 12323540

MOUNISH DAMERA- 12302154

1. Introduction

- **Background:** Briefly discuss the importance of high-lift devices in aviation, particularly slotted flaps.
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- **Objective:** Clearly state the goal of your experiment, which is to fabricate and test an NACA 23012 slotted airfoil to evaluate its aerodynamic performance.
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- **Scope:** Outline the specific aspects of the experiment, such as the range of angles of attack, wind tunnel conditions, and data analysis methods.

2. FABRICATION PROCESS :-

The fabrication of the slotted NACA 23012 airfoil model involved multiple steps to ensure high precision and aerodynamic accuracy. The key stages are as follows:

1. Airfoil Selection:

The NACA 23012 airfoil was chosen for its well-documented aerodynamic performance and compatibility with slotted modifications. The slotted design was specifically implemented to enhance lift characteristics by delaying flow separation.

2. Design and Modeling:

- The airfoil geometry was generated using **ANSYS** to ensure precision in 3D modeling.
- Slot dimensions and positions were optimized to achieve the desired aerodynamic effects, particularly for increased lift and reduced drag.

3. Material Selection:

The model was fabricated using **Polylactic Acid (PLA)**, a durable and lightweight material suitable for 3D printing. PLA's smooth finish after post-processing also ensures reduced surface roughness, enhancing aerodynamic accuracy.

4. CAD Design:

A detailed 3D CAD model was created in **Creo 11.0**, incorporating all geometric features, including the slot. The model was designed to fit the wind tunnel's test section while maintaining dimensional accuracy.

5. 3D Printing:

- The model was fabricated using a high-precision **3D printer**, layer by layer.
- Special care was taken to ensure dimensional fidelity, particularly for the slot geometry, which plays a critical role in the airfoil's aerodynamic performance.

6. Post-Processing:

- The airfoil was sanded using fine-grit **sandpaper** to eliminate surface irregularities caused by the printing process.
- A smooth surface finish was achieved, reducing boundary layer disturbances during wind tunnel testing.

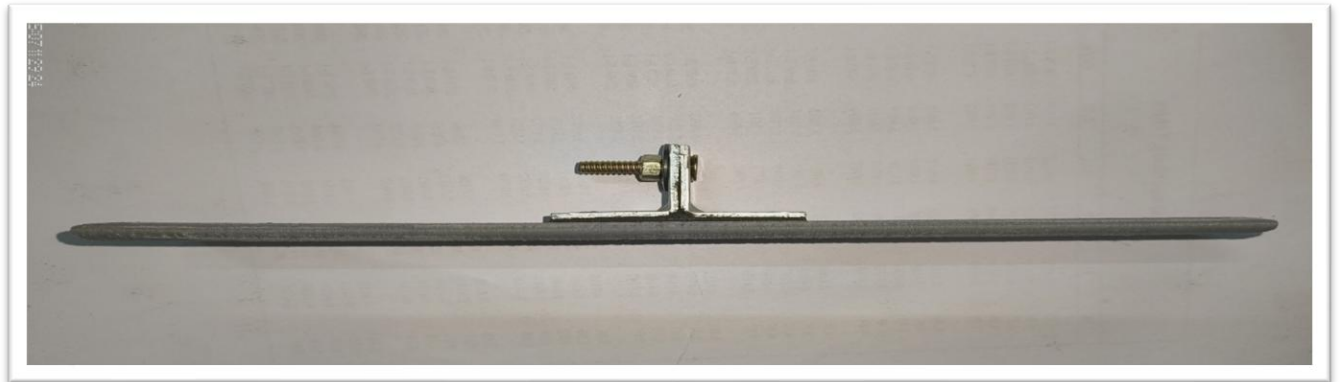
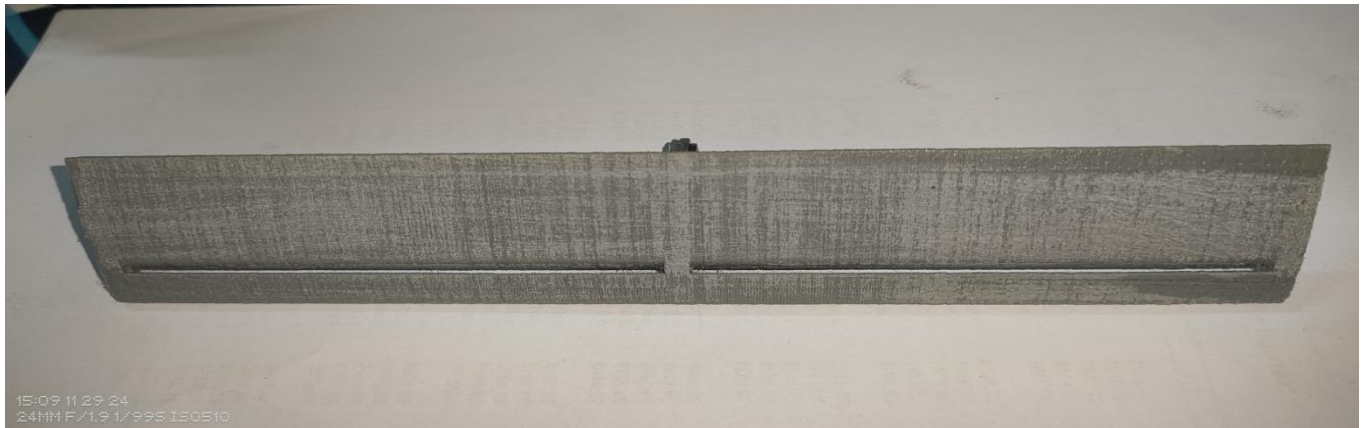
7. Slot Formation:

- The slot was formed with **high precision** tools, ensuring uniform width and alignment across the airfoil.
- Care was taken to avoid imperfections that could impact the flow behavior.

8. **Mounting Fixtures:**

Mounting points were incorporated into the model to facilitate secure placement in the wind tunnel. These fixtures were designed to minimize interference with the airflow.

STEP NO.	PROCESS	DETAILS
1	Airfoil Selection	NACA 23012
2	Design	Ansys (3D)
3	Material Selection	PLC
4	CAD design	Creo 11
5	Machining	3D printing
6	Finishing	Sand paper
7	Slot Formation	High precision



3. CALIBRATION PROCESS:-

- **Wind Tunnel Setup:**

- Set up the wind tunnel to achieve the desired flow conditions (velocity, lift, drag).
- Mount the airfoil securely in the test section, ensuring proper alignment and minimal blockage.

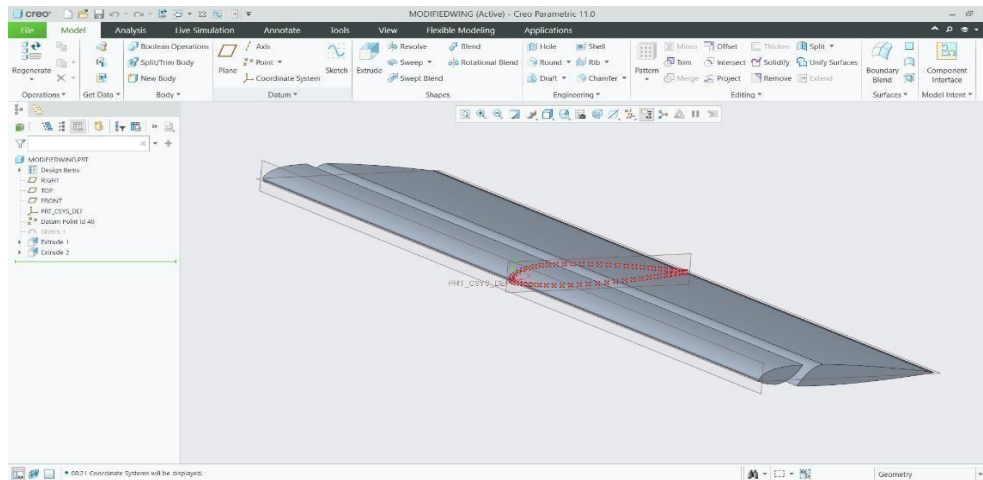
- **Calibration Procedure:**

- The calibration of wind tunnel is done to measure the tunnel speed, which can be measured through Pitot tube.
- The tunnel speed is the mean speed at the test section when the tunnel is empty.
- The tunnel speed is measured in terms of the difference between a total head and a static pressure reading.
- Calibration also ensures the uniformity of flow parameters in the region to be used for model testing.
- A Pitot tube is mounted just ahead of test section for this purpose.

- **Angle of Attack (AOA) Adjustment:**

- The setup was aligned to ensure AOA adjustments ranging from 0° to 5° with high accuracy.

4. SCALE CALCULATION



1. Reynolds Number Calculation:

The Reynolds number is calculated using:

$$Re = \rho \cdot V \cdot c / \mu$$

Where:

- $\rho = 1.225 \text{ kg/m}^3$.
- $V = 21.56 \text{ m/s}$.
- $c = 0.23 \text{ m}$
- $\mu = 1.81 \times 10^{-5} \text{ Pa}$.

Substituting:

$$Re = 1.225 \cdot 21.56 \cdot 0.23 / 1.81 \times 10^{-5} = 327,000$$

This Reynolds number is typical for low-speed wind tunnel tests.

2. Geometric Scaling Factor:

- If the model was scaled down (e.g., from a full-size airfoil to fit the wind tunnel), calculate the scaling factor. For example:

$$\text{Scale factor} = \text{Model Chord Length} / \text{Full-Scale Chord Length}$$

- If the **full-scale chord length** is 2.3 m then;

Given:

- Model Chord Length = 0.23 m
- Full-Scale Chord Length = 2.3 m

Substituting the values:

$$\text{Scale Factor} = 0.23 / 2.3 = 1:10$$

Alternatively, if we consider the reciprocal:

$$\text{Scale Factor} = \text{Full-Scale Chord Length} / \text{Model Chord Length}$$

$$\text{Scale factor} = 2.3 / 0.23 = 1:10$$

Thus, the scale factor is **1:10**, meaning the model is scaled down to 1/10th the size of the full-scale airfoil.

3. Force Scaling:

Forces such as lift and drag scale with the square of the scale factor:

$$\circ F_{\text{(real)}} = F_{\text{(model)}} \cdot (\text{Full-Scale Dimension} / \text{Model Scale Dimension})^2$$

Where:

- $F_{\text{(real)}}$: Force on the full-scale object
- $F_{\text{(model)}}$: Force measured on the model
- Full-Scale Dimension / Model Scale Dimension = Scale factor

This equation accounts for the proportional relationship between force and the square of the scale factor in aerodynamic testing.

5. TABULAR FORM:-

RPM :- 798

Pressure Velocity :- 23.7

Frequency :- 40Hz

AOA	Velocity (m/s)	Lift (N)	Drag (N)	Side force(N)	Pitching moment(Nm)	Rolling moment(Nm)	Yawing moment(Nm)
5	21.56	07.27	0.98	1.24	0.08	0.89	0.47



6. PROCEDURE :-

1. Check the wind tunnel for any loose parts. The wind tunnel was carefully checked for loose components or mechanical faults. Taking care of the robustness of the setup avoided creating any disturbance during the process of testing.
2. Adjust the Pitot tube at the mid-section height of the test section. The pitot tube was placed at the midpoint of the test section height. This ensured uniform and accurate measurements of the velocity of the flow during the experiment.
3. Run the tunnel at any RPM and note down the velocity reading. The wind tunnel was driven at a particular RPM; velocity values were taken. These data served as a base when calibrating the speed of flow.
4. Now run the tunnel at different RPM and note down the velocity reading. At different RPMs, the wind tunnel was operated; corresponding velocity values were noted. This helped to generate a correlation between RPMs and velocity of flow, very important in further analysis.
5. Measured lift, drag, and pressure at AOA increments of 2° from 0° to 5° .
6. Draw the calibration chart between Flow velocity and RPM. A calibration chart of flow velocity vs. RPM was plotted to provide an understanding of the wind tunnel's operation. The calibration chart served as a basis for subsequent results interpretation.



7. CONCLUSION :-

The study of the slotted NACA 23012 airfoil demonstrated significant improvements in aerodynamic performance, particularly in lift enhancement and drag reduction. Through meticulous fabrication and wind tunnel testing, the following observations were made:

1. The inclusion of a slot in the airfoil design successfully delayed flow separation, allowing for higher lift coefficients at increased angles of attack.
2. The wind tunnel experiments highlighted the advantages of the slotted airfoil over the standard configuration, particularly in terms of improved lift-to-drag ratio and stability under various flow conditions.
3. The fabrication process, including 3D printing and precision finishing, ensured an accurate representation of the design, contributing to reliable experimental results.

These findings underscore the potential of slotted airfoils for applications requiring enhanced lift characteristics, such as in high-lift devices for aircraft. Future work could explore variations in slot geometry and positions to further optimize performance or validate these results in higher Reynolds number flows.

